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DATA SHEETS

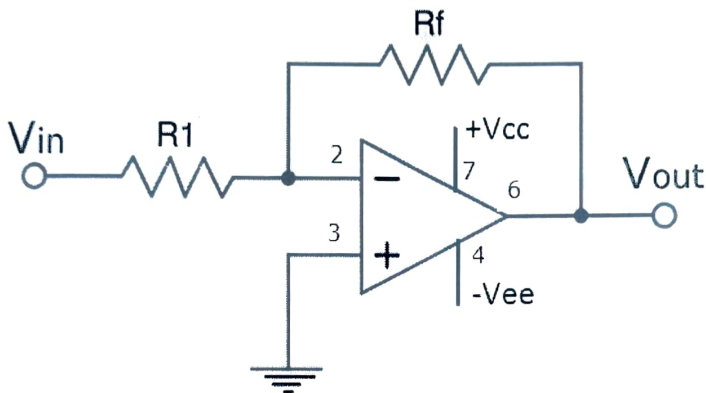
I.	LM 741 Op-amp IC	
II.	LM 555 Timer IC	

Experiment No: - 1

Aim: - To study op-amp 741 as inverting amplifier

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Function generator, CRO, Multimeter, Breadboard and connecting wires, Resistors.

Circuit diagram:-



Theory:-

Fig. shows close loop inverting amplifier. In this configuration, input signal is applied to inverting terminal of op-amp. The non-inverting terminal is grounded. The close loop voltage gain of (A_F) of inverting amplifier can be written as:

$$A_F = V_0/V_{in} = -R_F/R_1.$$

Where, A_F = Close loop voltage gain, V_0 =Output voltage, V_{in} =Input voltage, R_F =Feedback resistance, R_1 =Input resistance. [Write theory briefly from book].

Procedure:-

1. Connect the circuit as shown in Fig.
2. Connect dual DC supply to pin no. 7($+V_{cc}$) and pin no. 4 ($-V_{cc}$).
3. Connect AC input to inverting terminal of op-amp i.e. pin no. 2.
4. Connect output at pin no. 6 to CRO and observe inverted waveform.

5. Vary the frequency of input and measure output voltage V_0 .
6. Calculate gain (dB) and plot the frequency response on semi-log paper.

Observation Table: - $V_{in} = \underline{\hspace{2cm}}$, $R_F = \underline{\hspace{2cm}}$, $R_1 = \underline{\hspace{2cm}}$

Sr. No	Freq. 1kHz 100kHz -----	Output voltage V_0	Calculated gain $= -R_F/R_1$	Observed gain $= V_0/V_{in}$	Gain(dB) = $20\log$ (
1					
2					
-					

Result:-

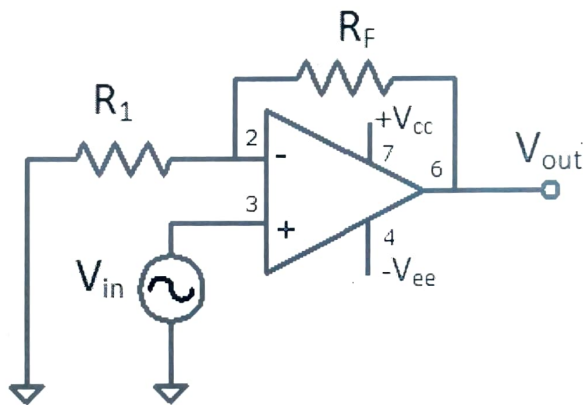
Precautions:-

Experiment No: - 2

Aim: - To study op-amp 741 as non-inverting amplifier

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Resistors, Function generator, CRO, Multimeter, Breadboard and connecting wires

Circuit diagram:-



Theory:-

Fig. shows close loop non-inverting amplifier. In this configuration, input signal is applied to non-inverting terminal of op-amp. The inverting terminal is grounded. The close loop voltage gain of (A_F) of non-inverting amplifier can be written as:

$$A_F = V_o/V_{in} = 1 + \frac{R_F}{R_1}$$

.Where, A_F = Close loop voltage gain, V_o =Output voltage, V_{in} =Input voltage, R_F =Feedback resistance, R_1 =Input resistance. [Write theory briefly from book].

Procedure:-

1. Connect the circuit as shown in Fig.
2. Connect dual DC supply to pin no. 7($+V_{cc}$) and pin no. 4 ($-V_{cc}$).

3. Connect AC input to non-inverting terminal of op-amp i.e. pin no. 3.
4. Connect output at pin no. 6 to CRO and observe waveform.
5. Vary the frequency of input and measure output voltage V_o .
6. Calculate gain (dB) and plot the frequency response on semi-log paper.

Observation Table: -

$$V_{in} = \underline{\hspace{2cm}}, R_F = \underline{\hspace{2cm}}, R_1 = \underline{\hspace{2cm}}$$

Sr. No	Freq. 1kHz 100kHz	-----	Output voltage V_o	Calculated gain = $1 + \frac{R_F}{R_1}$	Observed gain = V_o/V_{in}	Gain(dB) = $20\log\left(\frac{V_o}{V_{in}}\right)$
1						
2						
-						

Result:-

Precautions:-

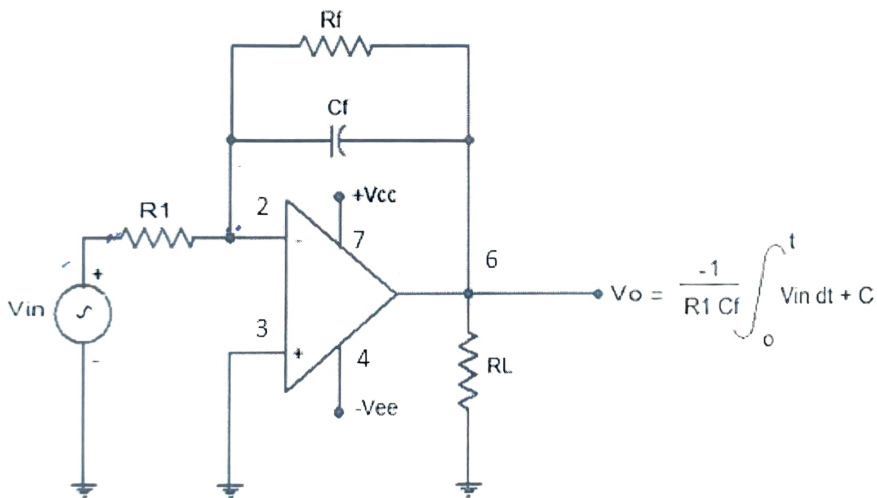
Experiment No: - 3

Aim: - To study op-amp 741 as integrator

Objectives: - (a) To study practical integrator circuit (b) To study frequency response of integrator.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Resistors, Function generator, CRO, Multimeter, Breadboard and connecting wires

Circuit diagram:-



Theory:-

The Figure.1 above shows an inverting integrator. The output is given

by $V_{out}(t) = \frac{-1}{R_1 C_F} \int_0^t V_{in}(t) dt.$

$R_1 C_F$ denotes time constant of the integration. The gain of the circuit decreases at the rate of -20dB/decade. Resistor R_F limits gain at low frequency and prevents it from becoming infinite for DC and low frequency signal. By limiting low frequency gain, the frequency response will flatten out and rolls down after critical frequency f_o at rate of -20dB/decade.

The gain limiting frequency $f_a = \frac{1}{2\pi R_F C_F}$. The unity gain frequency $f_b = \frac{1}{2\pi R_1 C_F}$.

Normally, between f_o and f_b the circuit acts as integrator and $f_o < f_b$. The input is integrated properly if time period T of signal is larger than or equal to $R_F C_F$.

Procedure:-

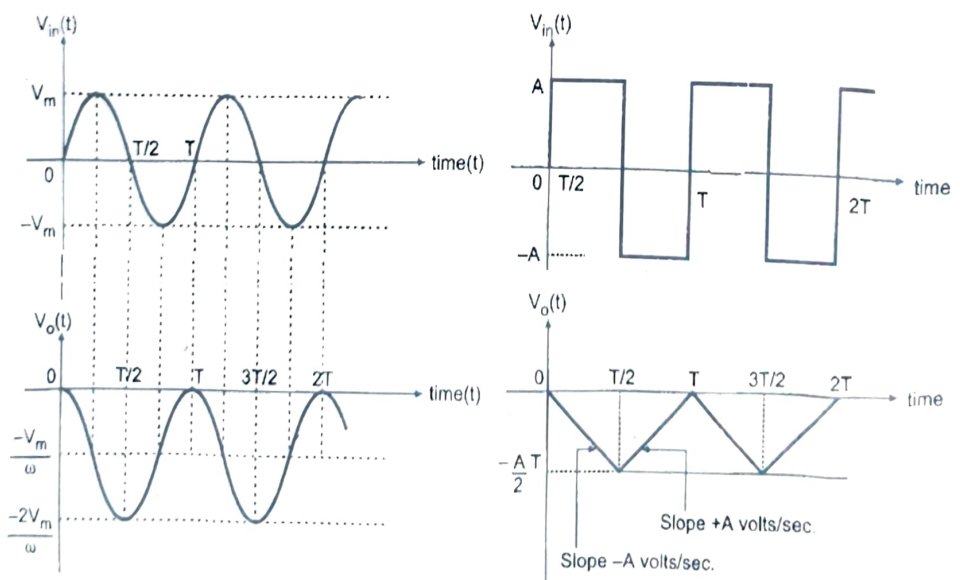
1. Connect the circuit as shown in Fig.
2. Connect dual DC supply to pin no. 7(+ V_{CC}) and pin no. 4 (- V_{CC}).
3. Connect AC input to inverting terminal of op-amp i.e. pin no. 2.
4. Connect output at pin no. 6 to CRO and observe integrated waveform.
5. Connect sine, square and triangular waveform as input and observe integrated output.
6. Vary the frequency of input and measure output voltage V_o .
7. Calculate gain (dB) and plot the frequency response on semi-log paper.

Observation Table: -

$V_{in} = \underline{\hspace{2cm}}$, $R_F = \underline{\hspace{2cm}}$, $R_1 = \underline{\hspace{2cm}}$, $C_F = \underline{\hspace{2cm}}$

Sr. No	Freq. 100Hz ---100kHz	Output voltage V_o	Observed gain = V_o/V_{in}	Gain(dB) = $20\log(V_o/V_{in})$
1				
2				

Model graph:-



Frequency Response:-

Plot the frequency response on semi-log paper. Indicate region of integration and compare observed f_a, f_b with calculated ones.

Result:-**Precautions:-**

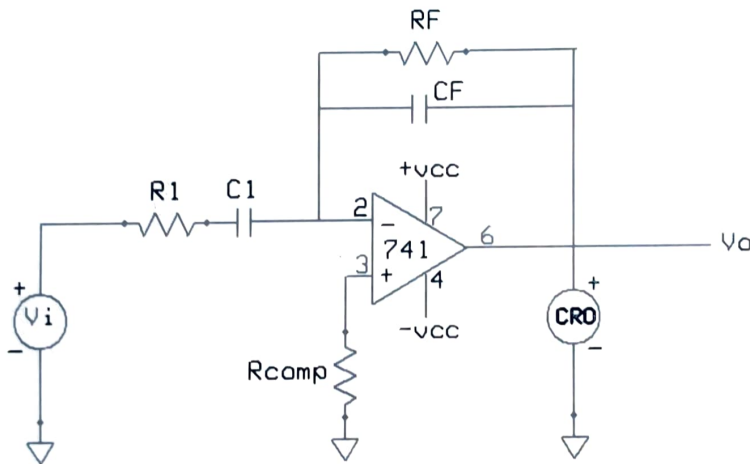
Experiment No: -4

Aim: - To study op-amp 741 as differentiator

Objectives: - (a) To study practical differentiator circuit (b) To study frequency response of differentiator.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Function generator, CRO, Multimeter, Breadboard and connecting wires , Resistors,Capacitors.

Circuit diagram:-



Theory:-

The Figure.1 above shows an inverting differentiator. The output is given by $V_{out}(t) = -R_F C_1 (dV_{in}(t)/dt)$. This makes circuit unstable. The gain of the circuit increases with the frequency at the rate of +20dB/decade. Feedback capacitor C_F connected across R_F limits the gain at high frequency. The input is differentiated properly if time period T of signal is larger than or equal to $R_F C_1$. The gain limiting frequency is $f_b = \frac{1}{2\pi R_1 C_1}$. The unity gain frequency is $f_a = \frac{1}{2\pi R_F C_1}$. and the unity-gain bandwidth is f_c . Differentiator can be designed by implementing following steps:

- Select f_o equal to highest frequency of input signal to be differentiated.
Assume $C_1 < 1 \mu F$, calculate R_F .
- Choose $f_b = 20 f_o$ and calculate R_1, C_F so that $R_1 C_1 = R_F C_F$.

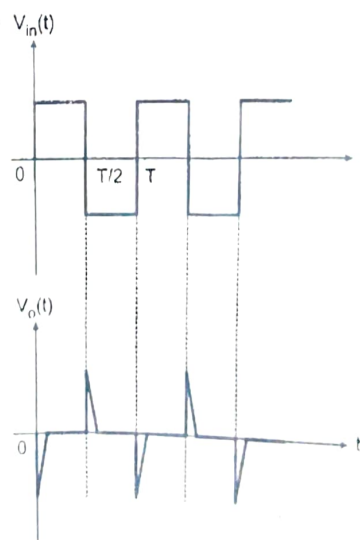
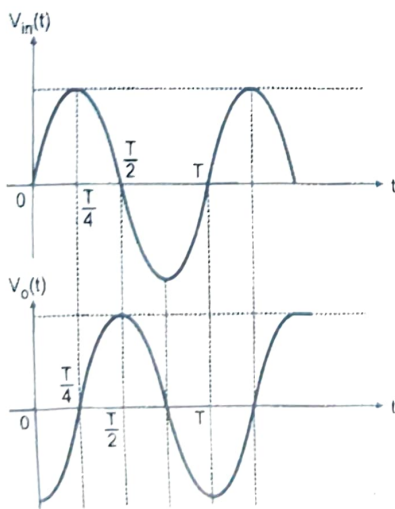
Procedure:-

1. Connect the circuit as shown in Fig.
2. Connect dual DC supply to pin no. 7(+ V_{CC}) and pin no. 4 (- V_{CC}).
3. Connect AC input to inverting terminal of op-amp i.e. pin no. 2.
4. Connect output at pin no. 6 to CRO and observe differentiated waveform.
5. Connect sine, square and triangular waveform as input and observe differentiated output.
6. Vary the frequency of input and measure output voltage V_o .
7. Calculate gain (dB) and plot the frequency response on semi-log paper.

Observation Table: - $V_{in} = \underline{\hspace{2cm}}$, $R_F = \underline{\hspace{2cm}}$, $R_1 = \underline{\hspace{2cm}}$, $C_F = \underline{\hspace{2cm}}$

Sr. No	Freq. 100Hz ---100kHz	Output voltage V_o	Observed gain = V_o/V_{in}	Gain(dB) = $20\log(V_o/V_{in})$
1				
2				

Model graph:-



Frequency Response:-

Plot the frequency response on semi-log paper. Indicate region of differentiation and compare observed f_a, f_b with calculated ones.

Result:-**Precautions:-**

Experiment No: - 5

Aim: - To study application of op-amp 741 as filter.

Objectives: - To study frequency response of (a) first order low pass butterworth filter (b) first order high pass butterworth filter.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Function generator, CRO, Multimeter, Breadboard and connecting wires, Resistors, Capacitors.

Circuit diagram and frequency response:-

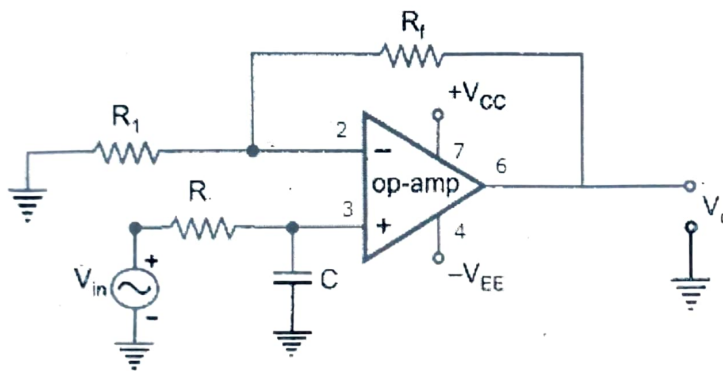


Fig.1 First order low pass butterworth filter

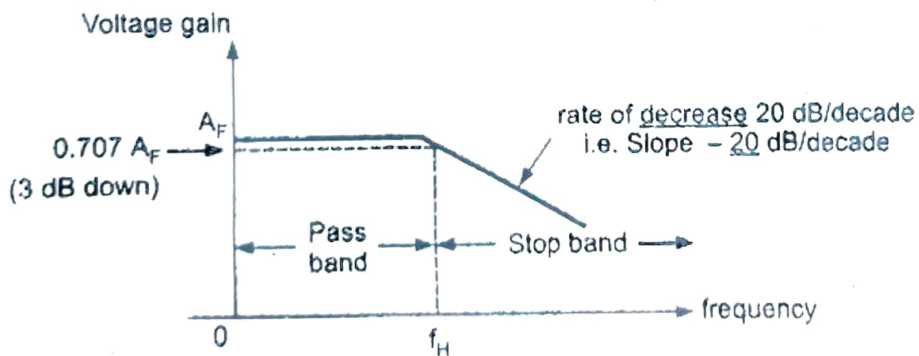


Fig. 2 Frequency response

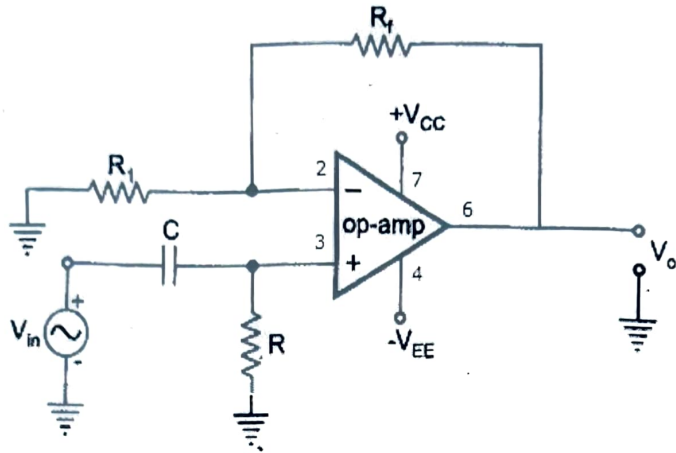


Fig.3 First order high pass Butterworth filter

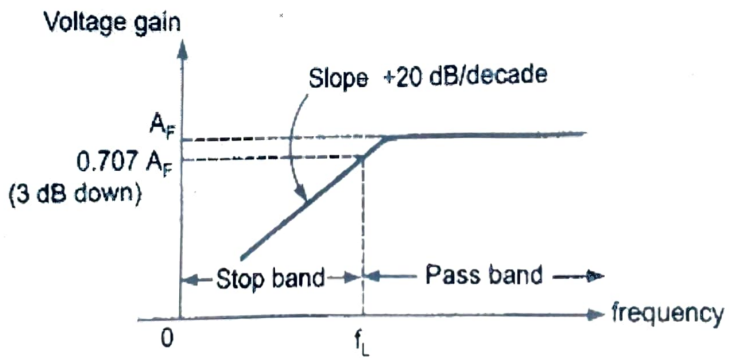


Fig. 4 Frequency response

Theory:-

Filter is a frequency selective circuit that passes a specified band of frequencies and blocks/ attenuates signals of frequencies outside this band. Depending upon the type of element used in their construction they are classified as:

Passive filter:-Elements used are resistors, capacitor, and inductors.

Active filter:-Employ transistor or op-amp in addition to passive elements.

First order Butterworth filters:-Key characteristic of Butterworth filter is that it has a flat pass band as well as stop band. The rate at which gain of filter changes

the stop band is determined by order of the filter. For first order filter it changes by 20 dB / decade.

Low pass filter: - It has a constant gain from 0 Hz to high cut-off frequency f_H . At f_H , gain is down by 3dB and for input frequency above f_H , gain decreases at the rate of 20 dB / decade in the stop band.

$$V_0/V_{in} = A_F / (1 + j(f/f_H)).$$

Where, V_0/V_{in} = voltage gain of filter as function of frequency, $A_F = 1 + \frac{R_F}{R_1}$ = pass band gain of filter. f = frequency of input, $f_H = \frac{1}{2\pi RC}$ = high cut-off frequency of the filter. At $f = f_H$, $V_0/V_{in} = 0.707 A_F$.

High pass filter: - This is obtained by interchanging the position of frequency determining resistor and capacitor in low pass filter. For $f > f_L$, It has a constant gain i.e. pass band. Stop band is for $0 < f < f_L$ and gain increases at the rate of 20 dB / decade in the stop band.

$$V_0/V_{in} = A_F [jf/f_L] / [1 + j(f/f_L)].$$

Where, V_0/V_{in} = voltage gain of filter as function of frequency, $A_F = 1 + \frac{R_F}{R_1}$ = pass band gain of filter. f = frequency of input, $f_L = \frac{1}{2\pi RC}$ = low cut-off frequency of the filter. At $f = f_L$, $V_0/V_{in} = 0.707 A_F$.

Procedure:-

1. Connect the circuit as shown in Fig.
2. Connect dual DC supply to pin no. 7(+V_{CC}) and pin no. 4 (-V_{CC}).
3. Connect AC input to non-inverting terminal of op-amp i.e. pin no. 3.
4. Connect output at pin no. 6 to CRO and measure it.
5. Vary the frequency of input and measure output voltage V_0 .
6. Calculate gain (dB) and plot the frequency response on semi-log paper.

Calculations (Low pass filter): - For specified high cut-off frequency f_H and pass band gain A_F , determine following:

$$R = \underline{\hspace{2cm}}, C = \underline{\hspace{2cm}}, R_F = \underline{\hspace{2cm}}, R_1 = \underline{\hspace{2cm}}$$

Observation Table (Low pass filter): -

Plot frequency response i.e. Gain (dB) vs. frequency on semi-log paper and determine observed high cut-off frequency f_H and compare with calculated ones.

Sr. No	Freq.(Hz)	Output voltage V_o	Observed gain $= V_o/V_{in}$	Gain(dB) = $20\log(V_o/V_{in})$
1				
2				
-				

Calculations (High pass filter): - For specified low cut-off frequency f_L and pass band gain A_F , determine following:

$$R = \text{____}, C = \text{____}, R_F = \text{____}, R_1 = \text{____}$$

Observation Table (High pass filter): -

Plot frequency response i.e. Gain (dB) vs. frequency on semi-log paper and determine observed low cut-off frequency f_L and compare with calculated ones.

Sr. No	Freq.(Hz)	Output voltage V_o	Observed gain $= V_o/V_{in}$	Gain(dB) = $20\log(V_o/V_{in})$
1				
2				
-				

Results:-

Precautions:

Experiment No: - 6

Aim: - To study op-amp 741 as (a) positive comparator (b) negative comparator.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Resistors, Function generator, CRO, Multimeter, Breadboard and connecting wires

Circuit diagram:-

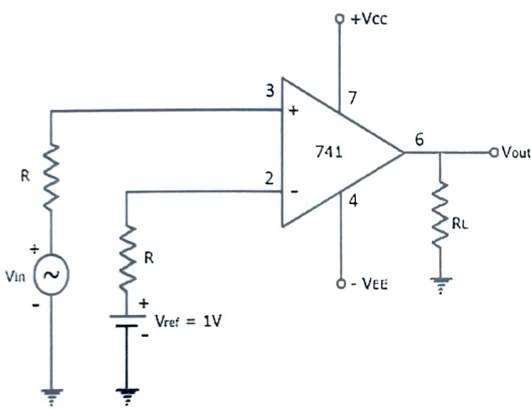


Fig.1 Non-Inverting Comparator Circuit

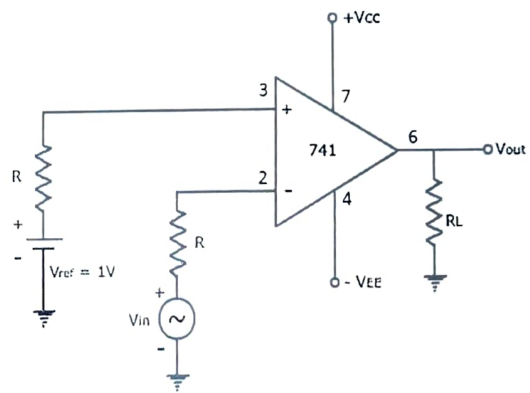


Fig.2 Inverting Comparator Circuit

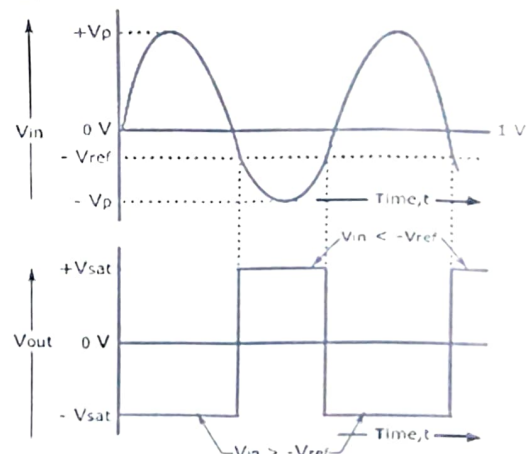
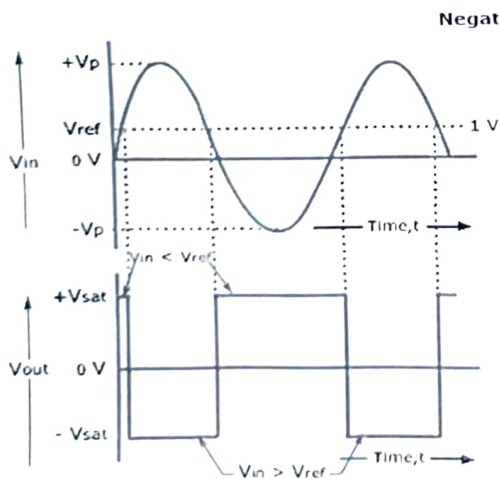
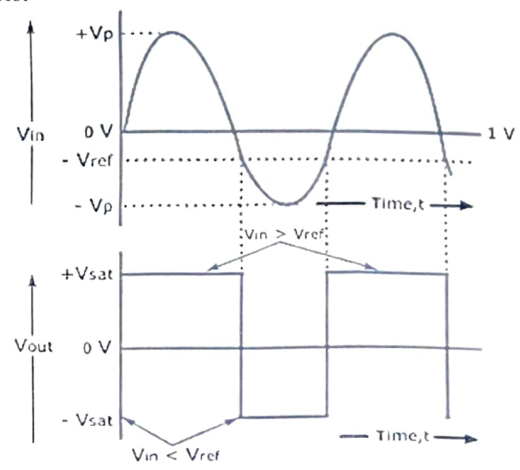
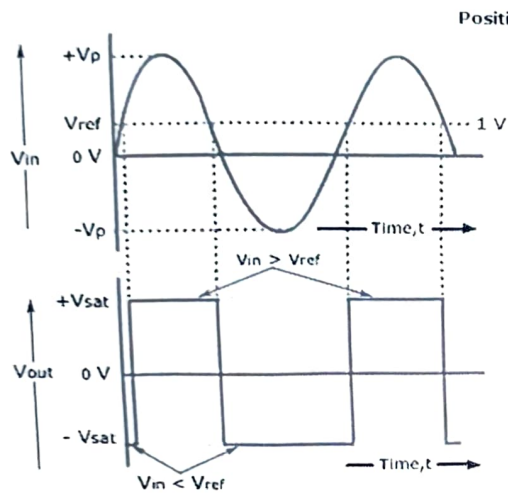
Theory: - A fixed reference voltage is applied to (-) input and time varying signal is applied to (+) input as shown in Fig.1. Because of this arrangement, the circuit is called positive or non-inverting comparator. When V_{in} is less than V_{ref} output is at $-V_{sat}$. On other hand when V_{in} is greater than V_{ref} , (+) input becomes positive with respect to (-) input, and output V_o goes to $+V_{sat}$. The comparator is also called voltage level detector because for a desired reference voltage V_{ref} , the voltage level of input V_{in} can be detected. For Fig. 2 write theory in your words.

Procedure:-

1. Connect the circuit as shown in Fig. 1
2. Connect dual DC supply to pin no. 7($+V_{CC}$) and pin no. 4 ($-V_{CC}$).
3. Connect AC input to non-inverting terminal of op-amp i.e. pin no. 3.
4. Connect reference voltage of +1 volt at inverting terminal.

5. Connect output at pin no. 6 to CRO and observe the waveform.
6. Plot input-output waveform on graph paper.
7. For inverting comparator, apply AC input to inverting terminal and reference voltage at non-inverting terminal. Repeat steps 1,2,5,6.

Model waveforms:-



Observation Table: - $R = 10\ k$

$V_{in} = \underline{\hspace{2cm}}$	$V_o = \underline{\hspace{2cm}}$	$-V_{sat} = \underline{\hspace{2cm}}$	$+V_{sat} = \underline{\hspace{2cm}}$
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Observe input-output waveform on CRO and plot it on graph paper for both positive and negative comparator.

Result:-

Precautions:-

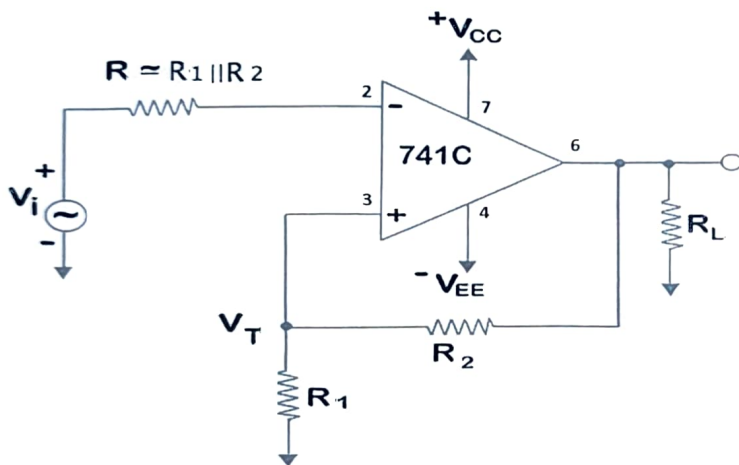
Experiment No: - 7

Aim: - To study op-amp 741 as Schmitt trigger.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Resistors, Function generator, CRO, Multimeter, Breadboard and connecting wires

Circuit diagram:-

SCHMITT TRIGGER USING OP - AMP 741C



Theory: - Fig above shows an inverting comparator with positive feedback. This circuit is known as Schmitt trigger and it converts irregular shaped waveform to square wave. The input voltage V_{in} triggers the output V_o every it exceeds certain voltage levels called upper threshold voltage V_{UTP} and lower threshold voltage V_{LTP} .

These threshold voltages are obtained by using the voltage divider R_1 - R_2 , where voltage across R_1 is fed back to (+) input. The threshold voltage is variable and depends upon value and polarity of output V_o .

The V_{UTP} and V_{LTP} are given by,

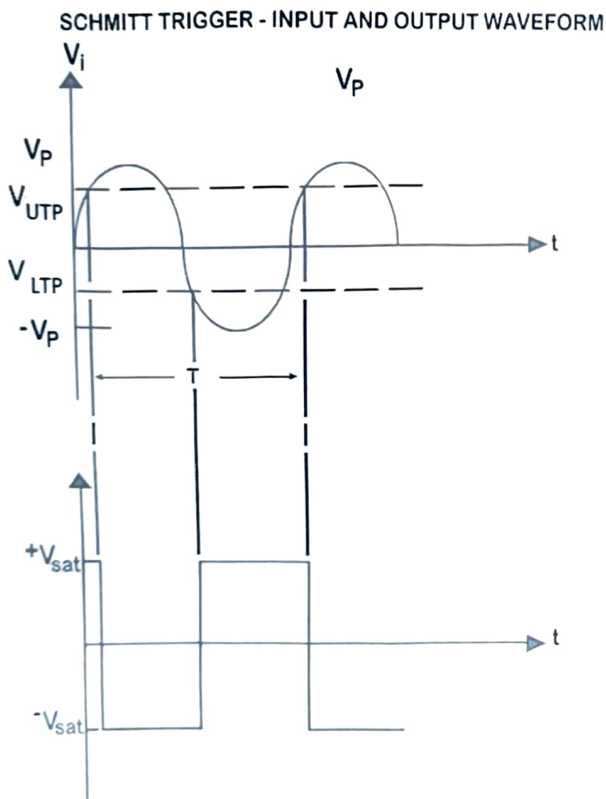
$$V_{UTP} = [R_1 / (R_1 + R_2)] * (V_{sat})$$

$$V_{LTP} = [R_1 / (R_1 + R_2)] * (-V_{sat})$$

Procedure:-

1. Connect the circuit as shown in Fig. 1
2. Connect dual DC supply to pin no. 7(+ V_{CC}) and pin no. 4 (- V_{CC}).
3. Connect AC input to inverting terminal of op-amp i.e. pin no 2.
4. Connect output at pin no. 6 to CRO and observe the square waveform.
5. Plot input-output waveform on graph paper.
6. Keep CRO in XY mode and observe hysteresis. Plot hysteresis on graph paper.

Model waveforms:-



Observations: - $R_2 = \underline{\hspace{2cm}}$, $R_1 = \underline{\hspace{2cm}}$,

$V_{in} = \underline{\hspace{2cm}}$	$V_o = \underline{\hspace{2cm}}$	$-V_{sat} = \underline{\hspace{2cm}}$	$+V_{sat} = \underline{\hspace{2cm}}$	$V_{UTP} = \underline{\hspace{2cm}}$	$V_{LTP} = \underline{\hspace{2cm}}$
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Observe input- output waveform on CRO and plot it on graph paper.

Result:-

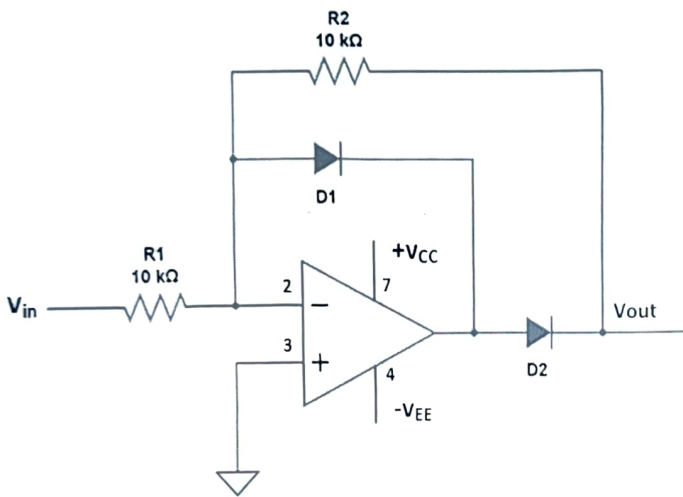
Precautions:-

Experiment No: - 8

Aim: - To study op-amp 741 as Half wave rectifier.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Resistors, Diodes, Function generator, CRO, Multimeter, Breadboard and connecting wires

Circuit diagram:-



Theory: - The major limitations of ordinary Si diode are that they cannot rectify voltage below 0.6V. A precision rectifier is a circuit capable of rectifying input of few mV. This circuit rectifies only one half of input signal and eliminates the other by bounding the output to zero volts. The input half cycle rectified can be either inverted or non-inverted. The circuit shown above is inverting amplifier which is converted into a HWR by adding two diodes.

When V_{in} is positive, diode D1 conducts causing op-amp to go negative by one diode drop ($\sim 0.6V$). Hence D2 is reversed biased and output voltage is zero and current flows through R_F .

For negative input V_{in} forces op-amp output to go positive and causes D2 to conduct. The circuit then acts like inverter for $R_F = R_1$ and $V_{out} = -V_{in}$.

Procedure:-

1. Connect the circuit as shown in Fig. 1
2. Connect dual DC supply to pin no. 7(+V_{CC}) and pin no. 4 (-V_{CC}).
3. Connect AC input to inverting terminal of op-amp i.e. pin no 2.
4. Connect output at pin no. 6 to CRO and observe the rectified waveform.
5. Plot input sine and half wave rectified output waveform on graph paper.
- 6.

Observations: - $R_2 = 10\text{ k}, R_1 = 10\text{ k},$

For $V_{in} = 2\text{ Volts p-p}$ ---- for 1st half cycle $V_{out} = \underline{\hspace{2cm}}?$

----for 2nd half cycle $V_{out} = \underline{\hspace{2cm}}?$

Observe input- output waveform on CRO and plot it on graph paper.

Result:-

Precautions:-

Experiment No: - 9

Aim: - To study op-amp 741 as Full wave rectifier.

Apparatus: - Op-Amp IC 741, Dual power supply $\pm 15V$, Function generator, CRO, Multimeter, Breadboard and connecting wires , Resistors , Diodes.

Circuit diagram:-

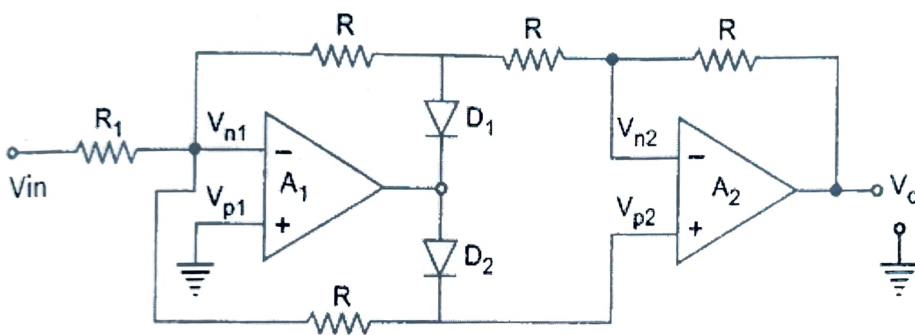


Fig. Full wave rectifier

Theory: - The circuit shown above is Full wave rectifier circuit. This circuit will give full wave rectified output for Sine wave signal as a input. For, positive half cycle, when $V_{in} > 0 V$, diode D_1 is on and D_2 is off. Both op-amplifiers A_1 and A_2 operates as inverting amplifier and it can be seen that $V_o = V_{in}$.

For negative half cycle, $V_{in} < 0 V$, diode D_1 is off and D_2 is on. Op-amp A_1 acts as an inverting amplifier and A_2 acts as a non-inverting amplifier. It can be seen that $V_o = -V_{in}$.

Procedure:-

1. Connect the circuit as shown in Fig. 1
2. Connect dual DC supply to pin no. 7($+V_{cc}$) and pin no. 4 ($-V_{cc}$).
3. Connect AC input to inverting terminal of op-amp i.e. pin no 2.
4. Connect output at pin no. 6 to CRO and observe the rectified waveform.
5. Plot input sine and full wave rectified output waveform on graph paper.

Observations: - $R = 10\text{ k}$,

For $V_{in} = 2\text{ Volts p-p}$ ----for 1st half cycle $V_{out} = \underline{\hspace{2cm}}?$

----for 2nd half cycle $V_{out} = \underline{\hspace{2cm}}?$

Observe input- output waveform on CRO and plot it on graph paper for half wave rectifier

Result:-

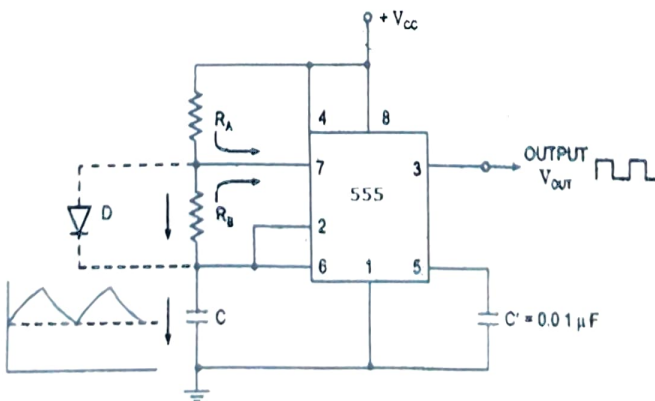
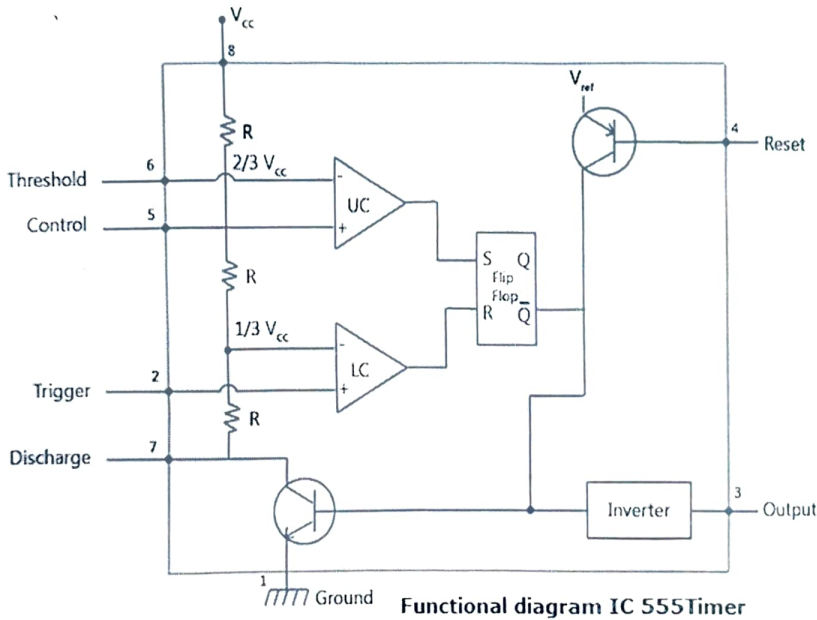
Precautions:-

Experiment No: - 10

Aim: - To study IC 555 as Astable Multivibrator.

Apparatus and Components: -, DC power supply, CRO, Multimeter, Breadboard and connecting wires, IC 555, Resistors, Capacitors, Diode

Functional and Circuit diagrams:-



Theory: -

IC 555 is a highly stable device for generating accurate time delays or oscillations. It can be used with supply voltage in the range of +5V to +18 V and can drive load up to 200mA. Figure 1 above shows internal functional diagram of IC555 and Figure 2 shows 555 configured for astable operation. Astable multivibrator doesn't require external trigger to change the state of the output. The time during which output is high is given by, $T_C = 0.69 (R_A + R_B)C$.

and time during which output is low is given by, $T_d = 0.69 R_B C$.

The total time period is given by, $T = T_C + T_d = 0.69 (R_A + 2R_B)C$ and % duty cycle = $(T_C/T) * 100$.

Procedure:-

1. Connect the circuit as shown in Fig. above.
2. Connect DC supply (+5 Volts) to pin no. 8(+V_{CC}) and pin no. 4 (Reset).
3. Connect output at pin no. 3 to CRO and observe the waveform.
4. Connect pin no. 2 to CRO and observe the waveform across the capacitor.

Calculations: - For specified frequency and duty cycle, determine $R_A = \underline{\hspace{2cm}}$, $R_B = \underline{\hspace{2cm}}$, $C = \underline{\hspace{2cm}}$

Observations: -

Observe output (V_{out}) waveform on CRO and plot it on graph paper. Find T_C , T_d , frequency and duty cycle of the waveform and compare with calculated ones.

Result:-

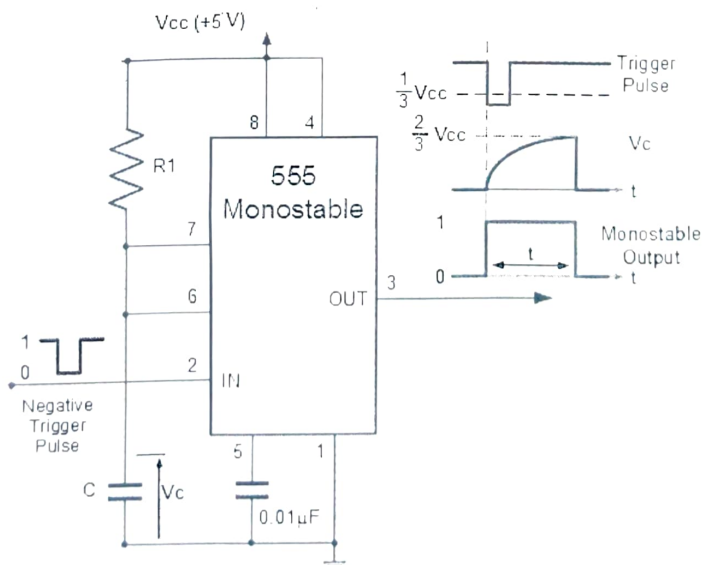
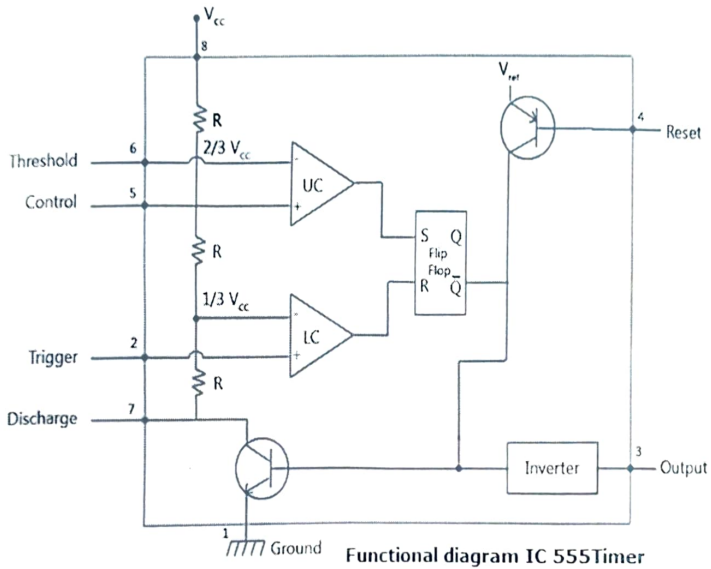
Precautions:-

Experiment No: - 11

Aim: - To study IC 555 as Monostable Multivibrator.

Apparatus and Components: -, DC power supply, CRO, Multimeter, Breadboard and connecting wires, IC 555, Resistors, and Capacitors.

Functional and Circuit diagrams:-



Theory: -

IC 555 is a highly stable device for generating accurate time delays or oscillations. It can be used with supply voltage in the range of +5V to +18 V and can drive load up to 200mA. Figure 1 above show internal functional diagram of IC55 and Figure 2 shows 555 configured for Monostable operation. Monostable multivibrator often called a one shot multivibrator is a pulse generating circuit. Duration of pulse is determined by R-C network connected externally to the IC 555. In standby state/stable state, the output of the circuit is at zero or logic low level. When external triggered is applied, output is forced to go high (+V_{cc}) and timing cycle now begins. After time period $T_{pulse} = 1.1 R_A C$, output returns to standby state (zero potential or logic low level) and it stays low until new trigger pulse is applied.

Procedure:-

1. Connect the circuit as shown in Fig. above.
2. Connect DC supply (+5 Volts) to pin no. 8(+V_{cc}) and pin no. 4 (Reset).
3. Connect function generator to pin no.2.
4. Connect output at pin no. 3 to CRO and observe the waveform.

Observations: - $R_A = \underline{\hspace{2cm}}$, $C = \underline{\hspace{2cm}}$

Observe output (V_{out}) waveform on CRO and plot it on graph paper. Find time period of output pulse and compare it with calculated ones.

Result:-

Precautions:-